Computer Architecture — Question Paper

1. CO1

a) Consider an IEEE754 representation where 9 bits are reserved for the exponent and 7 bits for the fraction. Now, find the following, and your final answer must be in decimal. Note: Consider three digits after the floating point while converting. [5]

i. Bias value.

ii. Smallest positive number.

iii. Largest positive number.

b) Given two numbers R = 0x7ABC and S = 0xBCDE in IEEE-754 floating-point representation, where 5-bits are assigned for the exponent, the rest of the bits are used for the sign and fraction fields. [4+2]

i. Perform R − S.

ii. Determine if the result you derived in b(i) has overflow, underflow, or none.

2. CO2

a) Determine whether the following statements are True or False. If false, then write the correct answer. [1.5+1.5+1.5]

i. Saved registers (x8–x9) and (x18–x27) must be restored to their original values before returning from a procedure if the procedure modifies them.

ii. The instruction sw x5, 0(x10) stores a word from memory into register x5.

iii. Consecutive word addresses in RISC-V main memory differ by 2 bytes.

b) Consider the following state of memory and registers (Table 1) at the start of execution. Now, for the instruction set being executed from Table 2, fill up the values of the registers in hexadecimal after the execution of the corresponding line on the right side of Table 2.

Table – 1: State of Memory and Registers

Address	Data (Hex)		Register	Value
0x2FBD	0xFF		x23	0xFF
0x2FBC	0xEE		x24	0x2FBB
0x2FBB	0xDD		x25	0x2FBA
0x2FBA	0xCC		x26	0x2FBC
...	...			
0x100	0xBB			
0xFF	0xAA			

Table – 2: Instruction Set

Memory Address	Instruction	Register Value
0x2FFE	lui x23, 0x102AA	x23 =
0x3002	lbu x24, 0(x25)	x24 =
0x3006	jal x1, f1	x1 =
0x300A	addi x24, x10, 0xF	x24 =
0x300E	beq x0, x0, end	
0x3012	f1: blt x25, x26, if	
0x3016	slli x24, x24, 63	
0x301A	beq x0, x0, exit	
0x301E	if: srli x24, x24, 62	
0x3022	exit: addi x10, x24, 0	x10 =
0x3026	jalr x0, 0(x1)	
0x302A	end:	

c) i. RISC-V uses a fixed instruction size (32 bits). Explain the disadvantage of using fixed-length instructions. [2]

ii. Explain the reason behind storing the jump offset as a jump immediate instead of the exact target address, even though the datapath needs more hardware components to compute the target address because of it. [2]

3. CO3

a) Consider the following instruction mix given below. In a Datapath scenario, answer the following questions: [2+2+2]

R-type	I-type (non-ld)	Load	Store	Branch
20%	15%	20%	15%	30%

i. For what fraction of all instruction register write signals are enabled and disabled?

ii. For what fraction of instruction's final outputs come directly from ALU?

iii. For what fraction of all instruction data does sign extension need?

b) Consider the following instruction sequence:

```
ld x6, 16(x7)
sub x6, x6, x8
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ld x6, 16(x7)
add x5, x9, x7

Suppose we modify the pipeline so that it has only one memory that handles both instructions and data. Explain the problem(s) that may arise while executing the above instruction sequence in the modified pipeline. Clearly identify the type of hazard introduced, and indicate during which instruction(s) the conflict occurs. [2+1]

c) [Figure: Fraction of a Full Datapath – showing Registers (Read register 1, Read register 2, Write register, Write data, Read data 1, Read data 2), ALUSrc mux, Imm Gen, ALU (with Zero and ALU result output), MemToReg mux, and Data memory (Address, Write data, Read data).]

When silicon chips are fabricated, defects in the material and manufacturing errors can result in defective circuits. Assume such a defect could be that one signal wire goes directly and always registers a logical 0. Considering this answer, the following questions: [3+3]

i. Write the names of two instructions that will fail to operate correctly if the MemToReg control signal is stuck at 0? Justify your answer.

ii. Write the names of two instructions that will fail to operate correctly if the ALUSrc control signal is stuck at 0? Justify your answer.

d) Consider the following RISC-V assembly code:

add x6, x6, x7
sub x5, x5, x6
sd x6, 16(x5)
ld x8, 16(x5)
or x12, x9, x6
sd x7, 24(x12)

Suppose we run the instruction set in a pipelined processor. Answer the following questions: [2+2.5+1]

i. Find the total number of data hazards.

ii. Draw a pipelined diagram to solve the Data hazard using stalling and forwarding.

iii. If the system has a 4s clock, find the total time needed to run the instruction set in the pipelined datapath.

For question 4, you must write your final answers in the question paper and show all the necessary calculations on the script.

4. CO4

a) You are designing a cache with 32 blocks, where each block can store 40 bits of data. For a memory address (111100111)■, find the index, tag, and offset bits. [3]

Tag bits =	Index bits =	Offset bits =

b) Given a direct-mapped cache with:

Cache size: 16 blocks
Block size: 16 bytes
Memory is byte-addressable with 12-bit addresses.

i. For the following memory addresses (in decimal): 160, 480, 160, 805, 278. Determine the cache index (binary) where each memory address will be mapped, and if we try to access the addresses 160, 480, 160, 805, 278 sequentially, determine hit/miss. Assume the cache is initially empty. [5]

Address (Decimal)	Cache Index (Binary)	Hit/Miss
160		
480		
160		
805		
278		

ii. Suppose the system has a 1 ns clock. It takes 3 clock cycles to determine if an access is a hit or a miss. It takes 2 ns to pass the data from the cache to the CPU, and it takes 5 ns to copy a block from the lower-level memory to cache. Determine the following: [4]

Hit Time =	____ ns
Miss Penalty =	____ ns
Miss Rate =	____ (Hint: Use the data from Question 4.b to find the answer.)
AMAT =	____ ns